# Couette flow

Wiki *Enhanced dissipation for passive scalars*

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Page team: to be named · Status: skeleton
Discuss on Zulip: #examples, topic "Couette flow".

*[Lead: what this example shows, in three lines.]*

## 1 Setting

*[The flow $u = (y, 0)$ on $\mathbb{T} \times \mathbb{R}$: the equation, the Fourier modes, what decays.]*

## 2 The result

**Theorem 1** (enhanced dissipation for Couette flow)**.** *[The statement.]*

*Idea. [The key mechanism in a few lines.]*

*Proof architecture. [The proof as a short chain of steps, and its diagram.]*

## 3 One example, every technique

Each proof below is written for this example only, to show the mechanism; the general results, with complete proofs, are in the part *Techniques and general results*.

### 3.1 Explicit solution

*Proof by the explicit solution. [The proof.]* □

### 3.2 Spectral methods

*Proof by spectral methods. [The proof.]* □

General theory: Spectral methods.

### 3.3 Hypocoercivity

*Proof by hypocoercivity. [The proof.]* □

General theory: Hypocoercivity.

### 3.4 Stochastic methods

*Proof by stochastic methods. [The proof.]* □

General theory: Stochastic methods.

### 3.5 Hörmander theory

*Proof by Hörmander theory. [The proof.]* □

General theory: Hörmander theory.

## 4 What this example teaches

*[What the example shows, and what the general theory keeps of it.]*

## References